05 — Dataset Strategy Document

Interpretable Machine Learning for Discovering Hidden Regulatory Switches in Non-Coding DNA

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1. Candidate Datasets

1.1 Priority Sources (Phase 1–2)

ENCODE Candidate Cis-Regulatory Elements (cCREs)

Property	Detail
Source	ENCODE portal (https://www.encodeproject.org/) and ENCODE SCREEN (https://screen.encodeproject.org/)
What it provides	Pre-classified candidate regulatory regions in the human genome: promoter-like signatures (PLS), proximal enhancer-like signatures (pELS), distal enhancer-like signatures (dELS), CTCF-only, DNase-H3K4me3
Genome assembly	hg38 (GRCh38)
Format	BED files downloadable from SCREEN or ENCODE portal
Size	~926,535 human cCREs (as of ENCODE phase 3); varies by filter
Why it matters	Provides ready-made, expert-curated labels for regulatory regions; eliminates need for raw signal processing
Licensing	ENCODE data use policy: freely available for research; requires citation

ENCODE DNase-seq / ATAC-seq Peak Files

Property	Detail

Source	ENCODE portal; individual experiment pages
What it provides	Cell-type-specific open chromatin peaks (regions of accessible DNA)
Format	BED / narrowPeak
Size	~50–200 MB per experiment (peaks); hundreds of experiments available
Why it matters	Enables cell-type-specific regulatory classification (a region is "regulatory in K562" if it has a DNase peak in K562)
Recommended experiments	Start with well-characterized cell types: K562 (ENCSR000EMT), GM12878 (ENCSR000EMU)

ENCODE Histone ChIP-seq Peak Files

Property	Detail
Source	ENCODE portal
What it provides	Peaks for histone marks associated with regulatory activity (H3K27ac = active enhancers; H3K4me3 = active promoters; H3K4me1 = poised enhancers)
Format	BED / narrowPeak
Why it matters	Distinguishes enhancer-like vs. promoter-like regions; enables element-type classification
Recommended experiments	H3K27ac and H3K4me3 in K562 and GM12878

Human Reference Genome (hg38)

Property	Detail
Source	UCSC Genome Browser (https://hgdownload.soe.ucsc.edu/goldenPath/hg38/bigZips/) or NCBI
What it provides	Full DNA sequence of the human genome
Format	FASTA (per-chromosome files recommended for memory efficiency)
Size	~3.1 GB total; ~250 MB per large chromosome
Why it matters	Required for extracting DNA sequences at annotated coordinates

JASPAR Motif Database

Property	Detail
Source	https://jaspar.elixir.no/

What it provides	Position frequency matrices (PFMs) for known transcription factor binding motifs
Format	MEME / PFM / JASPAR format
Size	~10 MB
Why it matters	Cross-referencing learned features against known biology; motif enrichment analysis

GENCODE Gene Annotations

Property	Detail
Source	https://www.gencodegenes.org/
What it provides	Gene coordinates, TSS positions, exon/intron boundaries
Format	GTF / GFF3
Size	~50 MB
Why it matters	Needed to exclude coding regions; identify promoter-proximal vs. distal regulatory regions

1.2 Secondary Sources (Phase 3+)

Dataset	What it provides	When to use
GTEx (https://gtexportal.org/)	Tissue-specific gene expression across 54 tissues	When linking regulatory predictions to expression
Roadmap Epigenomics	Epigenetic maps across 111 reference epigenomes	When expanding beyond ENCODE cell types
FANTOM5	CAGE-defined enhancers and promoters	Alternative enhancer labels for cross-validation
Single-cell ATAC-seq (e.g., from Satpathy et al.)	Cell-type-resolved chromatin accessibility	When exploring cell-type-specific regulation at finer resolution

1.3 Kaggle-Available Resources

Resource	Kaggle availability
AlphaGenome model	Available on Kaggle Models (for Stage 3 embedding extraction)
hg38 reference	Can be uploaded as a Kaggle dataset (per-chromosome FASTA)
ENCODE cCREs	Must be curated and uploaded as a Kaggle dataset
JASPAR PFMs	Small enough to include directly in notebook or dataset

2. Which Dataset to Start With

Recommended Starting Dataset: ENCODE cCREs (SCREEN V3)

Rationale:

1. **Pre-classified**: cCREs come with element-type annotations (PLS, pELS, dELS, CTCF-only), eliminating the need for raw signal processing.
2. **Well-curated**: Generated by the ENCODE consortium using standardized pipelines across multiple assays.
3. **Manageable size**: ~926K elements, but can be filtered to ~50K–200K per cell type for initial experiments.
4. **Directly downloadable**: Available as BED files from SCREEN.
5. **Widely cited**: Used in numerous computational biology studies; results are comparable to literature.

Starting Subset Recommendation

Filter	Value	Resulting size (approx.)
Assembly	hg38	Full set
Cell type	K562 (primary)	~150K–200K active cCREs
Element type	dELS + PLS (enhancer + promoter)	~100K–150K
Quality	High-confidence only (default SCREEN filters)	~80K–120K
Region length	200–2000 bp (filter extremes)	~70K–100K

This gives ~70K–100K positive regions — sufficient for training and evaluation.

3. Label Strategy

3.1 Binary Classification Labels

Label	Definition	Source
Positive (regulatory)	Region annotated as cCRE by ENCODE SCREEN with active signal in target cell type	ENCODE cCRE BED file, filtered by cell-type activity
Negative (non-regulatory)	Random non-coding region NOT annotated as any cCRE and NOT overlapping any ENCODE annotation	Generated by negative sampling pipeline

3.2 Multi-Class Labels (Phase 2)

Label	Definition	Source
PLS (Promoter-Like Signature)	cCRE with high H3K4me3 and high DNase signal near TSS	ENCODE SCREEN classification

pELS (Proximal Enhancer-Like)	cCRE with high H3K27ac, within 2 kb of TSS	ENCODE SCREEN classification
dELS (Distal Enhancer-Like)	cCRE with high H3K27ac, > 2 kb from TSS	ENCODE SCREEN classification
CTCF-only	cCRE with CTCF binding but no other marks	ENCODE SCREEN classification
Non-regulatory	Sampled negative region	Generated

3.3 Cell-Type-Specific Labels

A region may be: - Active in K562 but inactive in GM12878 → labeled differently per cell type. - Active in both → labeled positive in both. - The system trains separate models per cell type and compares feature importances.

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4. Positive / Negative Set Design

4.1 Positive Set Construction

```
ENCODE cCREs (BED)
    ■
    ■■■ Filter by cell-type activity (e.g., K562-active)
    ■■■ Filter by element type (e.g., dELS + PLS)
    ■■■ Filter by region length (200–2000 bp)
    ■■■ Remove regions overlapping GENCODE exons (exclude coding)
    ■■■ Resize to fixed length (e.g., center ± 500 bp = 1 kb)
    ■
    ■■■ → Positive set: ~50K–100K regions
```

4.2 Negative Set Construction

```
hg38 genome
    ■
    ■■■ Generate random non-overlapping 1 kb regions
    ■■■ Exclude: all ENCODE cCREs (any cell type)
    ■■■ Exclude: all GENCODE exons, UTRs, known promoters (±2 kb of TSS)
    ■■■ Exclude: ENCODE blacklist regions (problematic mappability)
    ■■■ Match: GC content distribution of positive set (±5%)
    ■■■ Match: chromosome distribution of positive set
    ■
    ■■■ → Negative set: 1× positive set size (default)
```

4.3 Positive-to-Negative Ratio

Ratio	When to use
1:1	Default for initial experiments; balanced training
1:3	Sensitivity analysis; tests classifier behavior under mild imbalance
1:5	Stress test; closer to genome-wide base rate

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5. Preprocessing Steps

5.1 Sequence Extraction Pipeline

Step	Action	Tool	Output
1	Download cCRE BED file	`wget` / ENCODE API	BED file
2	Filter BED by criteria (cell type, element type, length)	`pandas` / `pybedtools`	Filtered BED
3	Generate negative regions	Custom script (`negative_sampling.py`)	Negative BED
4	Merge positive + negative BED	`pandas`	Combined BED with labels
5	Extract DNA sequences from hg38 FASTA	`pysam` / `pyfaidx`	FASTA or NumPy arrays
6	One-hot encode sequences	Custom (`onehot.py`)	NumPy array (N × L × 4)
7	Compute k-mer features	Custom (`kmer.py`)	NumPy array (N × 4^k)
8	Compute GC content, dinucleotide freq	Custom (`gc_content.py`)	NumPy array
9	Apply chromosome-holdout split	Custom	Train/val/test indices
10	Cache to disk	`np.savez_compressed` / `parquet`	Versioned cache files

5.2 Sequence Representation

Representation	Shape	Size per sample (1 kb)	Used by
One-hot	(1000, 4)	4 KB	CNN
k-mer frequency (k=4)	(256,)	2 KB	Classical ML
k-mer frequency (k=5)	(1024,)	8 KB	Classical ML
k-mer frequency (k=6)	(4096,)	32 KB	Classical ML (sparse)
GC + dinucleotide	(17,)	<1 KB	All models (auxiliary)

5.3 Quality Control Checks

Check	Purpose	Action if failed
No N-bases > 10% of sequence	Ensures sequence quality	Exclude region
No overlap between positive and negative sets	Prevents label contamination	Re-generate negatives

No overlap between train/val/test regions	Prevents leakage	Verify chromosome assignment
GC distribution match (KS test p > 0.05)	Ensures negatives are composition-matched	Re-sample negatives
Balanced class distribution (within 1:1 to 1:5)	Prevents degenerate training	Resample or apply class weights
Sequence length consistency	All regions are exactly fixed length	Resize or exclude

6. Train / Validation / Test Split Policy

6.1 Chromosome-Holdout Split (Primary)

Split	Chromosomes	Approx. genome fraction
Train	chr1–chr14, chrX	~70%
Validation	chr15–chr18	~15%
Test	chr19–chr22	~15%

6.2 Rationale

- **Why chromosome-holdout?** Regulatory elements on the same chromosome may share regulatory landscapes, chromatin domains, or be in linkage disequilibrium. Splitting by chromosome eliminates within-chromosome leakage.
- **Why chr19–22 for test?** chr19 is the most gene-dense human chromosome, providing a challenging and distinct test distribution. chr22 was the first chromosome sequenced and is well-characterized.
- **Why chrX in training?** chrX has unique regulatory biology (X-inactivation); including it in training allows the model to learn from this diversity. If X-specific analysis is needed, hold out chrX separately.

6.3 Cross-Validation (Within Training Set)

- 5-fold stratified cross-validation on training chromosomes for hyperparameter tuning.
- Each fold is **stratified by class label**, not by chromosome (since chromosome-holdout is already enforced at the train/test boundary).

7. Leakage Risks

Risk	Description	Mitigation
Region overlap	Same genomic coordinates in train and test	Chromosome-holdout eliminates this
Nearby regulatory elements	Train and test regions from adjacent genomic positions	Chromosome-holdout; minimum 100 kb distance between nearest train/test regions (if within-chromosome split is ever used)
Negative set contamination	Negative regions that are actually regulatory but unannotated	Exclude all ENCODE annotations from negatives, not just target cell type

Feature leakage via GC content	If negatives are not GC-matched, GC alone becomes predictive	GC-matched negative sampling
Batch effects	Data from different ENCODE phases processed differently	Use only ENCODE phase 3 data with consistent processing
Label leakage through sequence length	If positives and negatives have different native lengths	Fixed-length regions (resize all to 1 kb)

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8. Ethics and Licensing Considerations

8.1 Data Licensing

Dataset	License / Policy	Obligation
ENCODE	ENCODE Data Use Policy (https://www.encodeproject.org/help/data-use-policy/)	Free for research use; cite ENCODE consortium publications
hg38 reference genome	Public domain (NCBI/UCSC)	No restrictions
JASPAR	CC BY 4.0	Attribute JASPAR in publications
GENCODE	Freely available	Cite GENCODE
GTEx (future)	dbGaP controlled access for individual-level data; summary data publicly available	Use only summary/aggregate data unless dbGaP access is obtained

8.2 Ethical Considerations

Consideration	Status
Human subjects	Not applicable — no individual-level data; reference genome and population-level annotations only
IRB approval	Not required for this data
Personally identifiable information	None present in reference genome or ENCODE peak files
Dual-use risk	Low — regulatory prediction is basic research, not weaponizable
Bias	The reference genome (hg38) is derived primarily from a few individuals; regulatory annotations are biased toward well-studied cell types. This limits generalizability to under-represented populations and tissues. Acknowledge in publications.
Clinical misuse	Explicitly state that results are computational predictions, not clinical diagnostics. No claims about disease risk, treatment, or individual genome interpretation.

8.3 Responsible Reporting

- Frame all results as "candidate regulatory patterns" or "computational predictions."

- Do not claim that models have "discovered" regulatory switches — they have identified statistical associations.
- Report limitations prominently.
- Do not overstate generalizability beyond tested cell types and conditions.

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9. Data Versioning

9.1 Versioning Policy

Item	Version strategy
ENCODE cCRE download	Tag by download date and ENCODE release version (e.g., `cCRE_v3_2026-06-07`)
hg38 reference	Use UCSC hg38 with specific patch level (e.g., `GRCh38.p14`)
JASPAR	Tag by JASPAR release (e.g., `JASPAR2024`)
Processed datasets	Semantic version (e.g., `v0.1.0`) with changelog
Feature caches	Keyed by dataset version + feature config hash

9.2 Reproducibility Contract

For any experiment result, the following must be recoverable: 1. Exact dataset version used. 2. Exact feature configuration. 3. Exact data split (chromosome assignment). 4. Random seed. 5. Model hyperparameters. 6. Software versions (requirements.txt / conda environment).

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10. Data Budget (Kaggle Constraints)

Data item	Size on disk	Fits in Kaggle?
Curated cCRE BED (K562 + GM12878)	~50 MB	✓
hg38 per-chromosome FASTA (chr1–22, X)	~3.1 GB	✓ (as Kaggle dataset)
Processed sequences (100K regions × 1 kb, one-hot)	~400 MB	✓
k-mer features (100K × 4096, sparse)	~200 MB	✓
JASPAR PFMs	~10 MB	✓
GENCODE GTF	~50 MB	✓
Total estimated	**~4–5 GB**	**✓ (within 20 GB Kaggle limit)**

End of Document 05 — Dataset Strategy
Next: 06 — Modeling Roadmap